

What Are Kubernetes Deployment Strategies?

Deployment strategies define **how new versions of an application are rolled out** to users, ensuring minimal downtime and maximum availability.

Kubernetes supports multiple strategies; two of the most used are:

1. **Recreate**
2. **Rolling Update (default)**



Recreate vs Rolling Update: Theoretical Differences

Feature	Recreate Strategy	Rolling Update Strategy
Definition	Terminates all old pods before starting new ones	Gradually replaces old pods with new ones
Downtime	Yes – full downtime during transition	No or minimal – continuous service availability
Resource Usage	Lower – only one set of pods at a time	Slightly higher – both old and new pods run during update
Use Case	Suitable for major changes (e.g., DB schema change)	Ideal for production environments with live traffic
Rollback Speed	Fast – just redeploy previous version	Slower – gradually rolls back
Strategy Keyword	strategy: type: Recreate	strategy: type: RollingUpdate (default)

Practical Example with YAML Files

We'll create a deployment named my-app using both strategies for a sample image (e.g., nginx).

1. Rolling Update Strategy (default)

- rolling-update-deployment.yaml:

vim rolling-update-deployment.yaml

```
controlplane:~$ vim rolling-update-deployment.yaml
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-app
spec:
  replicas: 3
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
      maxSurge: 1
  selector:
    matchLabels:
      app: my-app
  template:
    metadata:
      labels:
        app: my-app
    spec:
      containers:
        - name: nginx
          image: nginx:1.19
          ports:
            - containerPort: 80
```

Update the Deployment:

- **kubectl apply -f rolling-update-deployment.yaml**

```
controlplane:~$ kubectl apply -f rolling-update-deployment.yaml
deployment.apps/my-app created
```

- **Kubectl get pods**

```
controlplane:~$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
my-app-6d55d95df9-62vcr	1/1	Running	0	18s
my-app-6d55d95df9-cx5tx	1/1	Running	0	18s
my-app-6d55d95df9-vpgwj	1/1	Running	0	18s

- **kubectl describe deployments.apps my-app**

```
controlplane:~$ kubectl describe deployments.apps my-app
Name: my-app
Namespace: default
CreationTimestamp: Tue, 27 May 2025 17:30:38 +0000
Labels: <none>
Annotations: deployment.kubernetes.io/revision: 1
Selector: app=my-app
Replicas: 3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType: RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 1 max unavailable, 1 max surge
Pod Template:
  Labels: app=my-app
  Containers:
    nginx:
      Image: nginx:1.19
      Port: 80/TCP
      Host Port: 0/TCP
      Environment: <none>
      Mounts: <none>
      Volumes: <none>
      Node-Selectors: <none>
```

- **kubectl set image deployment/my-app nginx=nginx:1.20**

```
controlplane:~$ kubectl set image deployment/my-app nginx=nginx:1.20
deployment.apps/my-app image updated
```

Kubectl get pods -w

```
controlplane:~$ kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
my-app-6d55d95df9-q5tjx             1/1     Running   0           11s
my-app-6d55d95df9-qppb6             1/1     Running   0           11s
my-app-6d55d95df9-szmq6             1/1     Running   0           11s
my-app-5c7c957bc-nwn7q              0/1     Pending   0           0s
my-app-5c7c957bc-nwn7q              0/1     Pending   0           1s
my-app-6d55d95df9-szmq6             1/1     Terminating 0          17s
my-app-5c7c957bc-nwn7q              0/1     ContainerCreating 0          1s
my-app-5c7c957bc-csx5r              0/1     Pending    0           0s
my-app-5c7c957bc-csx5r              0/1     Pending    0           0s
my-app-5c7c957bc-csx5r              0/1     ContainerCreating 0          0s
my-app-6d55d95df9-szmq6             1/1     Terminating 0          17s
my-app-6d55d95df9-szmq6             0/1     Completed   0          17s
my-app-5c7c957bc-nwn7q              0/1     ContainerCreating 0          1s
my-app-5c7c957bc-csx5r              0/1     ContainerCreating 0          0s
my-app-6d55d95df9-szmq6             0/1     Completed   0          17s
my-app-6d55d95df9-szmq6             0/1     Completed   0          17s
my-app-5c7c957bc-nwn7q              1/1     Running     0           2s
my-app-5c7c957bc-csx5r              1/1     Running     0           1s
my-app-6d55d95df9-q5tjx             1/1     Terminating 0          18s
my-app-5c7c957bc-58752              0/1     Pending    0           1s
my-app-5c7c957bc-58752              0/1     Pending    0           1s
my-app-6d55d95df9-qppb6             1/1     Terminating 0          19s
my-app-5c7c957bc-58752              0/1     ContainerCreating 0          1s
my-app-6d55d95df9-q5tjx             1/1     Terminating 0          19s
my-app-6d55d95df9-q5tjx             0/1     Completed   0          19s
my-app-6d55d95df9-qppb6             1/1     Terminating 0          19s
my-app-6d55d95df9-qppb6             0/1     Completed   0          19s
my-app-5c7c957bc-58752              0/1     ContainerCreating 0          1s
my-app-6d55d95df9-qppb6             0/1     Completed   0          19s
my-app-6d55d95df9-qppb6             0/1     Completed   0          19s
my-app-5c7c957bc-58752              1/1     Running     0           1s
my-app-6d55d95df9-q5tjx             0/1     Completed   0          19s
my-app-6d55d95df9-q5tjx             0/1     Completed   0          19s
```

What Happens:

- 1 pod (maxUnavailable: 1) is taken down.
- 1 new pod (maxSurge: 1) is created with the new image.
- Gradual replacement ensures service is always available.

2. Recreate Strategy

- recreate-deployment.yaml:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-app
spec:
  replicas: 3
  strategy:
    type: Recreate
  selector:
    matchLabels:
      app: my-app
  template:
    metadata:
      labels:
        app: my-app
    spec:
      containers:
      - name: nginx
        image: nginx:1.19
        ports:
        - containerPort: 80
```

Update the Deployment:

- `kubectl apply -f recreate-deployment.yaml`

```
controlplane:~$ kubectl apply -f recreate-deployment.yaml
deployment.apps/my-app created
```

- **KubectI get pods**

```
controlplane:~$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
my-app-6d55d95df9-d2cf4	1/1	Running	0	27s
my-app-6d55d95df9-l62np	1/1	Running	0	27s
my-app-6d55d95df9-nl6dl	1/1	Running	0	27s

- **kubectI set image deployment/my-app nginx=nginx:1.20**

```
controlplane:~$ kubectl set image deployment/my-app nginx=nginx:1.20
deployment.apps/my-app image updated
```

```
controlplane:~$ kubectl get pods -w
```

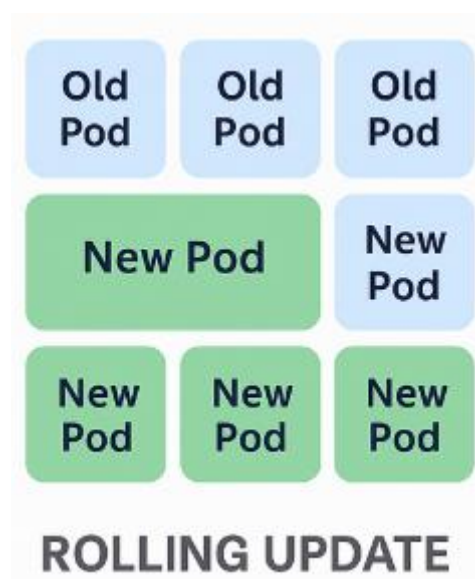
NAME	READY	STATUS	RESTARTS	AGE
my-app-6d55d95df9-d2cf4	1/1	Running	0	91s
my-app-6d55d95df9-l62np	1/1	Running	0	91s
my-app-6d55d95df9-nl6dl	1/1	Running	0	91s
my-app-6d55d95df9-l62np	1/1	Terminating	0	100s
my-app-6d55d95df9-nl6dl	1/1	Terminating	0	100s
my-app-6d55d95df9-d2cf4	1/1	Terminating	0	100s
my-app-6d55d95df9-l62np	1/1	Terminating	0	100s
my-app-6d55d95df9-d2cf4	1/1	Terminating	0	100s
my-app-6d55d95df9-nl6dl	1/1	Terminating	0	100s
my-app-6d55d95df9-l62np	0/1	Completed	0	100s
my-app-6d55d95df9-nl6dl	0/1	Completed	0	100s
my-app-6d55d95df9-d2cf4	0/1	Completed	0	100s
my-app-6d55d95df9-nl6dl	0/1	Completed	0	101s
my-app-6d55d95df9-nl6dl	0/1	Completed	0	101s
my-app-6d55d95df9-d2cf4	0/1	Completed	0	101s
my-app-6d55d95df9-d2cf4	0/1	Completed	0	101s
my-app-6d55d95df9-l62np	0/1	Completed	0	101s
my-app-6d55d95df9-l62np	0/1	Completed	0	101s
my-app-5c7c957bc-mtxk4	0/1	Pending	0	0s
my-app-5c7c957bc-mtxk4	0/1	Pending	0	0s
my-app-5c7c957bc-rqppl	0/1	Pending	0	0s
my-app-5c7c957bc-spgwg	0/1	Pending	0	0s
my-app-5c7c957bc-rqppl	0/1	Pending	0	0s
my-app-5c7c957bc-spgwg	0/1	Pending	0	0s
my-app-5c7c957bc-mtxk4	0/1	ContainerCreating	0	0s
my-app-5c7c957bc-rqppl	0/1	ContainerCreating	0	1s
my-app-5c7c957bc-spgwg	0/1	ContainerCreating	0	1s
my-app-5c7c957bc-mtxk4	0/1	ContainerCreating	0	1s
my-app-5c7c957bc-spgwg	0/1	ContainerCreating	0	1s
my-app-5c7c957bc-rqppl	0/1	ContainerCreating	0	1s
my-app-5c7c957bc-mtxk4	1/1	Running	0	2s
my-app-5c7c957bc-spgwg	1/1	Running	0	2s
my-app-5c7c957bc-rqppl	1/1	Running	0	2s

🔄 What Happens:

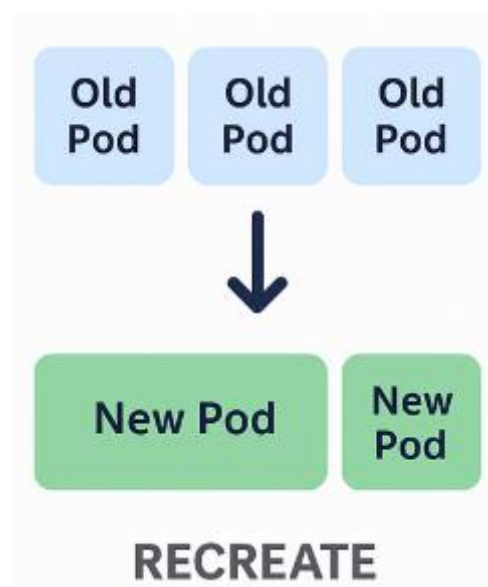
- All old pods are killed **first**.
- Only after termination, new pods with the updated image are started.
- Users experience **downtime** during the switch.

Visual Summary

Rolling Update:



Recreate:



When to Use Which?

☒ Rolling Update (Recommended for Most Cases):

- Continuous availability required
- Web apps, APIs, microservices

☒ Recreate:

- Major app overhauls
- DB-dependent services
- Avoid compatibility issues between old & new versions



KUBERNETES THOUGHTS

Before choosing a deployment strategy, evaluate your application's requirements and constraints.

